

Lab 3: Monitoring, Logging & DevSecOps

1. Set up monitoring using Prometheus

Set up monitoring using Prometheus and visualize metrics using Grafana dashboards for a target application or system.

```
Last login: Fri Oct 2 01:26:16 on tty000
anushkaberlia@Anushkas-MacBook-Air-2 ~ % mkdir -p ~/monitoring-lab/app && cd ~/monitoring-lab
anushkaberlia@Anushkas-MacBook-Air-2 monitoring-lab % cat > app/app.py << 'EOF'
import random, time
from prometheus_client import start_http_server, Counter, Histogram, Gauge

REQUESTS = Counter('app_requests_total', 'Total requests', ['endpoint', 'status'])
LATENCY = Histogram('app_request_latency_seconds', 'Request latency in seconds')
IN_PROGRESS = Gauge('app_in_progress_requests', 'Requests currently in progress')

start_http_server(8000) # metrics exposed at :8000/metrics

while True:
    IN_PROGRESS.inc()
    with LATENCY.time():
        time.sleep(random.uniform(0.05, 0.6))
        endpoint = random.choice(['/home', '/login', '/search'])
        status = random.choice(['200', '500'], weights=[90, 10])[0]
        REQUESTS.labels(endpoint, status).inc()
        IN_PROGRESS.dec()
    time.sleep(random.uniform(0.1, 0.5))
EOF

cat > app/Dockerfile << 'EOF'
FROM python:3.12-slim
RUN pip install --no-cache-dir prometheus_client
COPY app.py /app.py
CMD ["python", "-u", "/app.py"]
EOF
anushkaberlia@Anushkas-MacBook-Air-2 monitoring-lab % cat > prometheus.yml << 'EOF'
global:
  scrape_interval: 5s

scrape_configs:
  - job_name: prometheus
    static_configs:
      - targets: ['prometheus:9090']

  - job_name: node-exporter
    static_configs:
      - targets: ['node-exporter:9100']

  - job_name: sample-app
    static_configs:
      - targets: ['sample-app:8000']
EOF
anushkaberlia@Anushkas-MacBook-Air-2 monitoring-lab % >....
- ./prometheus.yml/etc/prometheus/prometheus.yml
ports:
  - "9090:9090"

node-exporter:
  image: prom/node-exporter:latest
  container_name: node-exporter
  ports:
    - "9100:9100"

sample-app:
  build: ./app
  container_name: sample-app
```

```
ports:
  - "9100:9100"

sample-app:
  build: ./app
  container_name: sample-app
  ports:
    - "8000:8000"

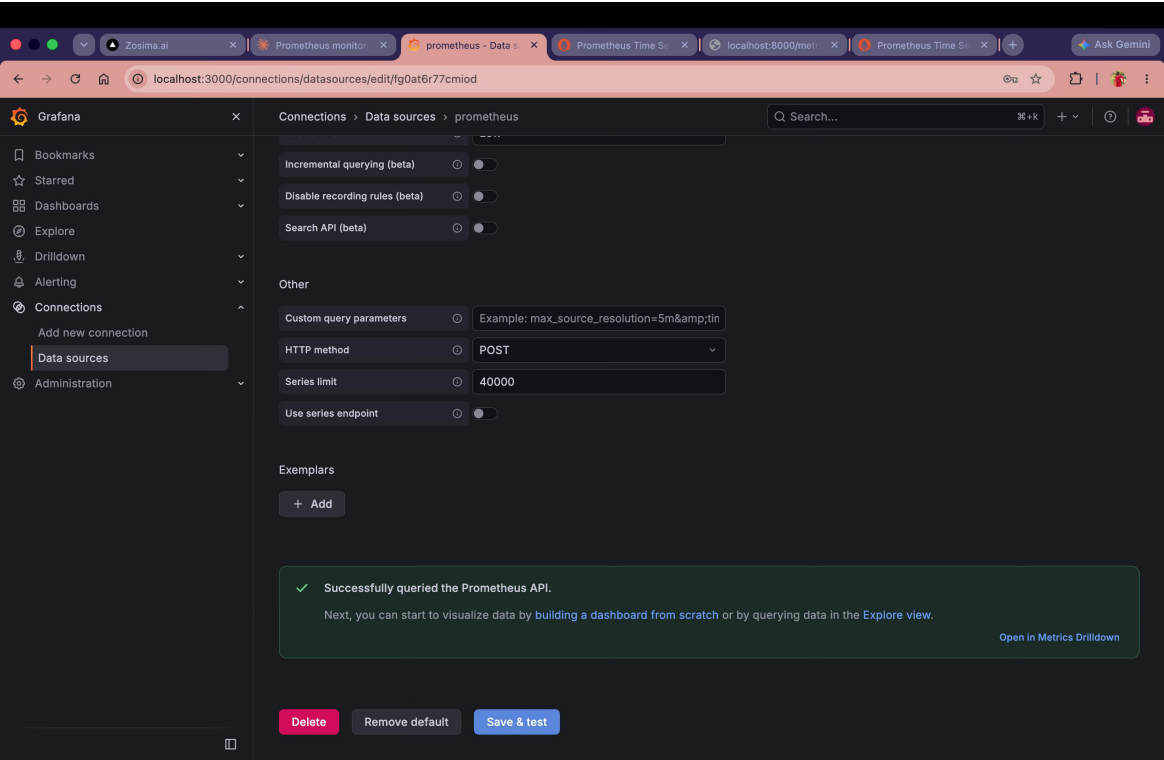
grafana:
  image: grafana/grafana:latest
  container_name: grafana
  ports:
    - "3000:3000"
  volumes:
    - grafana-data:/var/lib/grafana
  depends_on:
    - prometheus

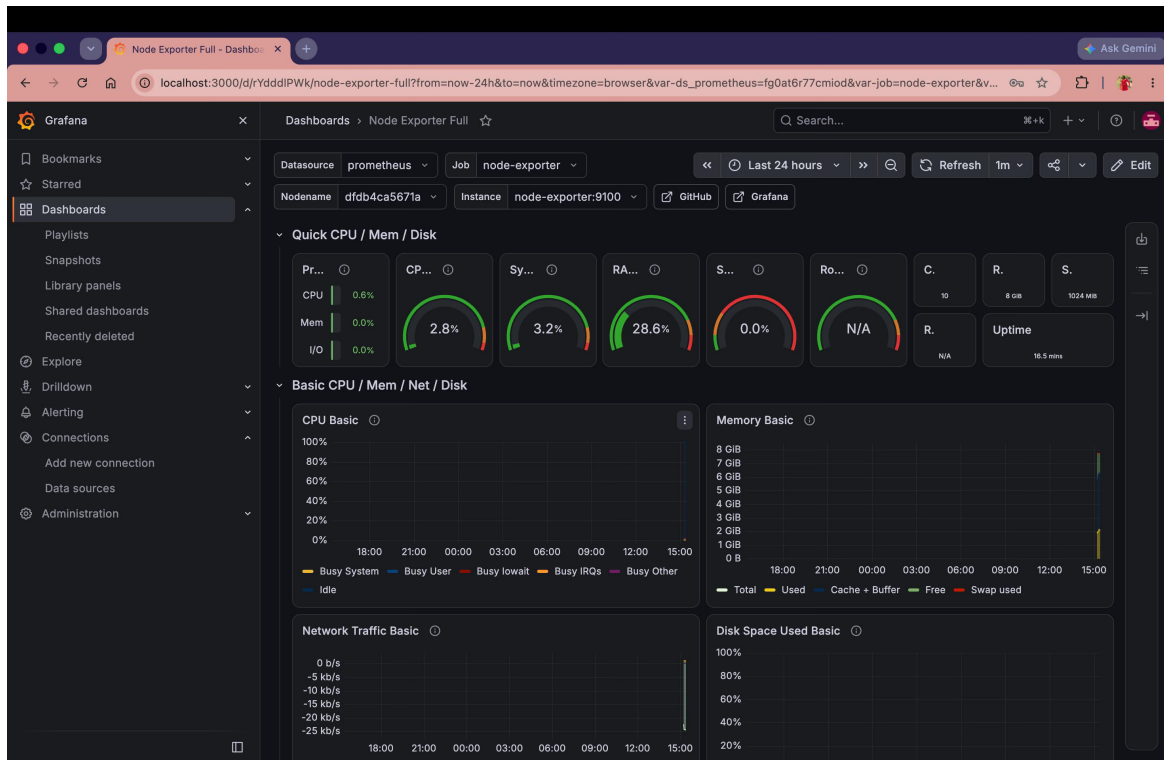
volumes:
  grafana-data:
EOF
anushkaberlia@Anushkas-MacBook-Air-2 monitoring-lab % docker compose up -d --build
docker compose ps
[+] up 24/24
✓ Image prom/node-exporter:latest Pulled 15.8s
✓ Image prom/prometheus:latest Pulled 25.4s
✓ Image grafana/grafana:latest Pulled 29.5s
[+] Building 12.1s (11/11) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 560B 0.0s
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 160B 0.0s
=> [internal] load metadata for docker.io/library/python:3.12-slim 4.3s
=> [auth] library/python:pull token for registry-1.docker.io 0.0s
=> [internal] load dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/3] FROM docker.io/library/python:3.12-slim@sha256:82188f5d322dd89ffc9e552442c25ac08543dfdbc456693a5599624f 4.9s
=> => resolve docker.io/library/python:3.12-slim@sha256:82188f5d322dd89ffc9e552442c25ac08543dfdbc456693a5599624f 0.0s
=> sha256:aad712613b67883876e7bdcdd78687aa7f5cc935cadcb0412abf686baee2c 248B / 248B 0.4s
=> sha256:58a64aacd115959e7a3ac9351be739ec8de44d5689b13c1445fc072db18079d2343d 12.0MB / 12.0MB 1.5s
=> sha256:bd52add1124e0215f0397d7f29ec8de44d5689b13c1445fc072db18079d2343d 4.61MB / 4.61MB 2.1s
=> sha256:bd36565c8fdeba8f03a5fc3b4ac619ca085ced32e9e9da856d95912f5f18f47b 30.19MB / 30.19MB 4.1s
=> extracting sha256:bd36565c8fdeba8f03a5fc3b4ac619ca085ced32e9e9da856d95912f5f18f47b 0.5s
=> extracting sha256:bd52add1124e0215f0397d7f29ec8de44d5689b13c1445fc072db18079d2343d 0.1s
=> extracting sha256:58a64aacd115959e7a3ac9351be739ec8de44d5689b13c1445fc072db18079d2343d 0.2s
=> extracting sha256:aad712613b67883876e7bdcdd78687aa7f5cc935cadcb0412abf686baee2c 0.0s
=> [internal] load build context 0.0s
=> => transferring context: 777B 0.0s
=> [2/3] RUN pip install --no-cache-dir prometheus_client 2.4s
=> [3/3] COPY app.py /app.py 0.0s
=> exporting to image 0.3s
=> => exporting layers 0.2s
=> => exporting manifest sha256:921f8cce87709135c3bb2ed9e85b9d5542456ee2f381589fc59650157cd173ce 0.0s
=> => exporting config sha256:821f8b68aa4a738c432bac583e222776147b740480f06b2587c5da99baa9ed2 0.0s
=> exporting attestation manifest sha256:9937e0b0920774cd39169624bc75b3c7f42d33cd34493222e2f6a771f804b 0.0s
[+] up 31/31ing manifest list sha256:703a73c521ca1438bc7092919444b682eb56c568c858da1965aaa3e4f311b 0.0s
=> Image prom/node-exporter:latest Pulled 15.8s
✓ Image prom/prometheus:latest Pulled 25.4s
✓ Image grafana/grafana:latest Pulled 29.5s
```

```
- "3000:3000"
volumes:
  - grafana-data:/var/lib/grafana
depends_on:
  - prometheus

volumes:
  grafana-data:
    grafana-data:
      anushkaberlia@Anushkas-MacBook-Air-2 monitoring-lab % docker compose up -d --build
      docker compose ps
      [+] up 24/24
      ✓ Image prom/node-exporter:latest Pulled 15.8s
      ✓ Image prom/prometheus:latest Pulled 25.4s
      ✓ Image grafana/grafana:latest Pulled 29.5s
      [+] Building 12.1s (11/11) FINISHED
      => [internal] load local bake definitions 0.0s
      => => reading from stdin 560B 0.0s
      => [internal] load build definition from Dockerfile 0.0s
      => => transferring dockerfile: 168B 0.0s
      => [internal] load metadata for docker.io/library/python:3.12-slim 4.2s
      => [auth] library/python:pull token for registry-1.docker.io 0.0s
      => [internal] load .dockerignore 0.0s
      => => transferring context: 2B 0.0s
      => [1/3] FROM docker.io/library/python:3.12-slim@sha256:82108f6d322dd89f1c9e552442c25ac08543dfdbc456693a5599624f 0.9s
      => => resolve docker.io/library/python:3.12-slim@sha256:82108f6d322dd89f1c9e552442c25ac08543dfdbc456693a5599624f 0.0s
      => => sha256:8add712613b07e8387e67bcdcd78607aa7f5cc935caddb0b412abf68bbaee2c 248B / 248B 0.4s
      => => sha256:58a544acd15b9797a3ac9c351be73bacf56d96fc49a7d1d04f2f8a587c6098 12.60MB / 12.60MB 1.8s
      => => sha256:b852add1124a21f839707799a8d64c4d58993c1c445fcd020b189792343a 4.61MB / 4.61MB 2.1s
      => => sha256:bd36565c8fdebafe03af3c3b4ace610ca885ced32e9e9da858d95912f5f18f47b 30.19MB / 30.19MB 4.1s
      => => extracting sha256:bd36565c8fdebafe03af3c3b4ace610ca885ced32e9e9da858d95912f5f18f47b 0.5s
      => => extracting sha256:bd52add1124a21f839707799a8d64c4d58993c1c445fcd020b189792343a 0.1s
      => => extracting sha256:58a544acd15b9797a3ac9c351be73bacf56d96fc49a7d1d04f2f8a587c6098 0.2s
      => => extracting sha256:8add712613b07e8387e67bcdcd78607aa7f5cc935caddb0b412abf68bbaee2c 0.0s
      => [internal] load build context 0.0s
      => => transferring context: 777B 0.0s
      => [2/3] RUN pip install --no-cache-dir prometheus_client 2.4s
      => [3/3] COPY app.py /app.py 0.0s
      => exporting to image 0.3s
      => => exporting layers 0.2s
      => => exporting manifest sha256:921f8cce87f99135c3bb2ed9e85b9d5542455ee2f3015997c59650157cd173ce 0.0s
      => => exporting config sha256:a2f8b68aae4a738c432bacb83e22277614f7b40a80f06b2587c5dad98aa9ed2 0.0s
      => => exporting attestation manifest sha256:9937e0b892b77ac539f8962c4bcf75b3c7f42d23cd346931221a2f6a771f8d48 0.0s
      [+] up 31/31ing manifest list sha256:783b736521ca1438bc7d92919464b62eb85ec5658c858da1965aa3e4f311b 0.0s
      ✓ Image prom/node-exporter:latest Pulled 15.8s
      ✓ Image prom/prometheus:latest Pulled 25.4s
      ✓ Image grafana/grafana:latest Pulled 29.5s
      ✓ Image monitoring-lab-sample-app Built 12.1s
      ✓ Network monitoring-lab_default Created 0.0s
      ✓ Volume monitoring-lab_grafana-data Created 0.0s
      ✓ Container sample-app Started 0.2s
      ✓ Container prometheus Started 0.3s
      ✓ Container node-exporter Started 0.1s
      ✓ Container grafana Started 0.3s
      NAME IMAGE COMMAND SERVICE CREATED STATUS PORTS
      grafana grafana/grafana:latest "/bin/sh" grafana Less than a second ago Up Less than a second 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
      node-exporter prom/node-exporter:latest "/bin/node_exporter" node-exporter Less than a second ago Up Less than a second 0.0.0.0:9100->9100/tcp, [::]:9100->9100/tcp
      prometheus prom/prometheus:latest "/bin/prometheus --c..." prometheus Less than a second ago Up Less than a second 0.0.0.0:9090->9090/tcp, [::]:9090->9090/tcp
      sample-app monitoring-lab-sample-app "python -u /app.py" sample-app Less than a second ago Up Less than a second 0.0.0.0:8000->8000/tcp, [::]:8000->8000/tcp
      anushkaberlia@Anushkas-MacBook-Air-2 monitoring-lab %
```

Connect Grafana to Prometheus



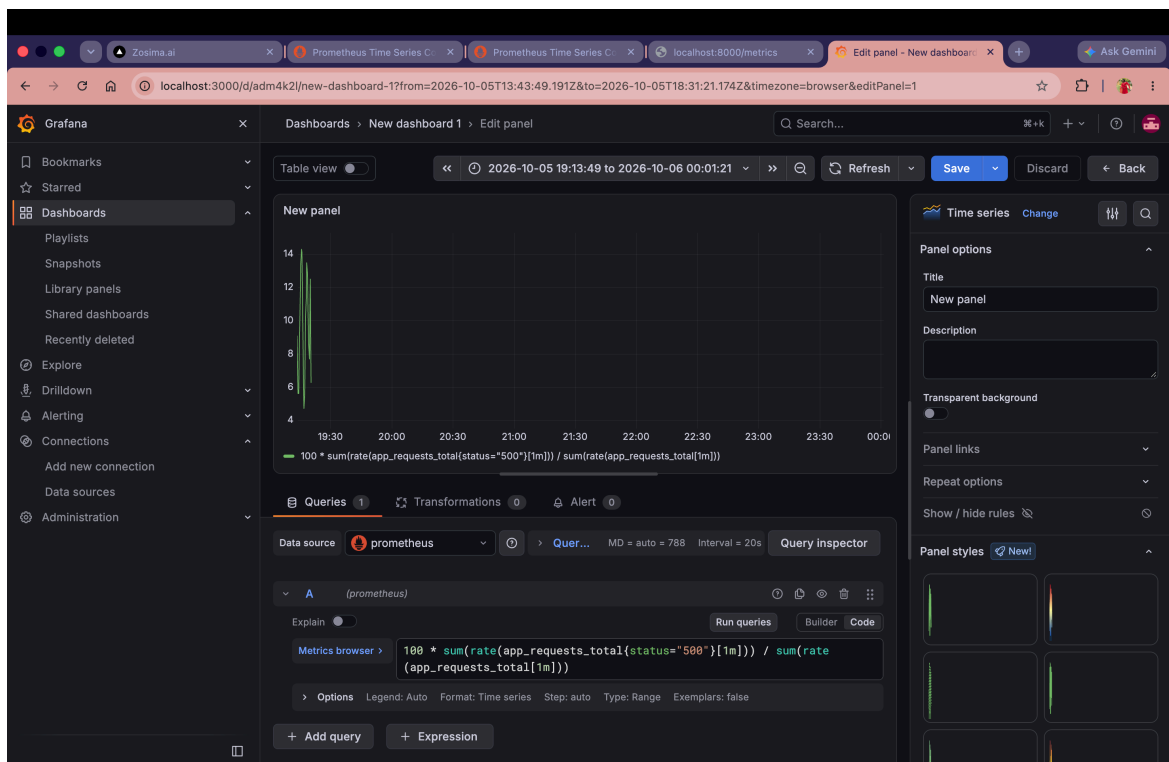
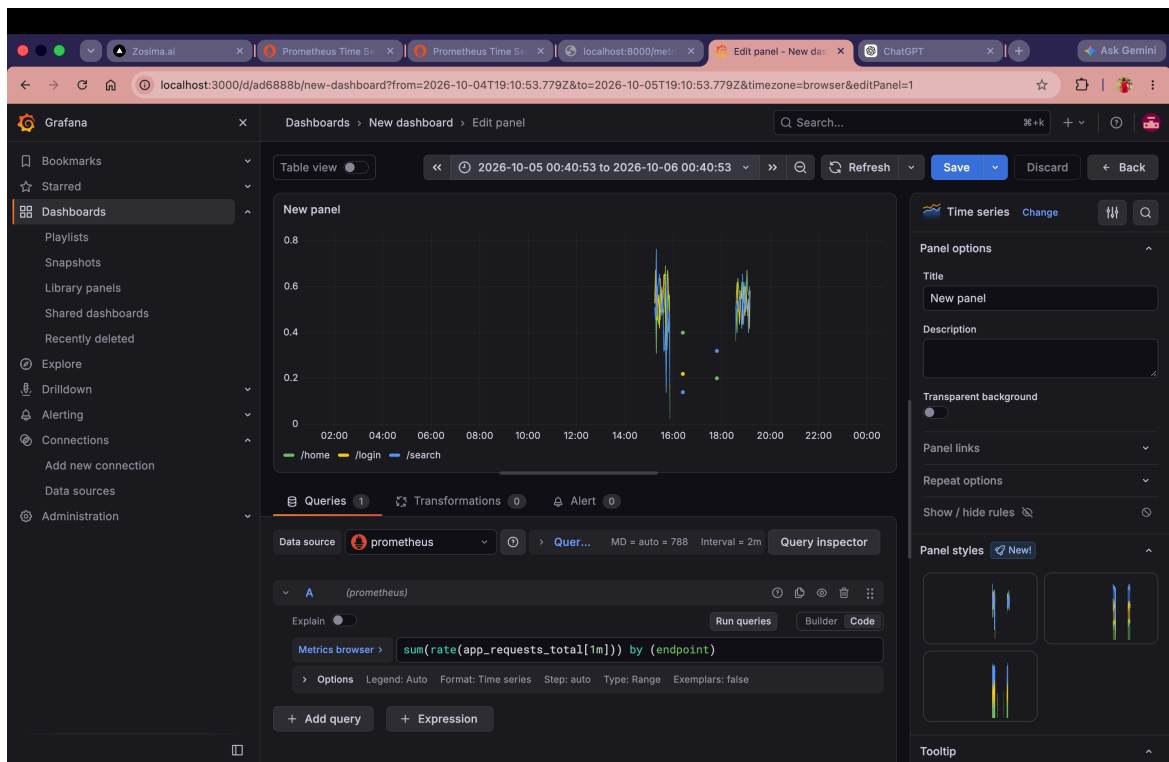


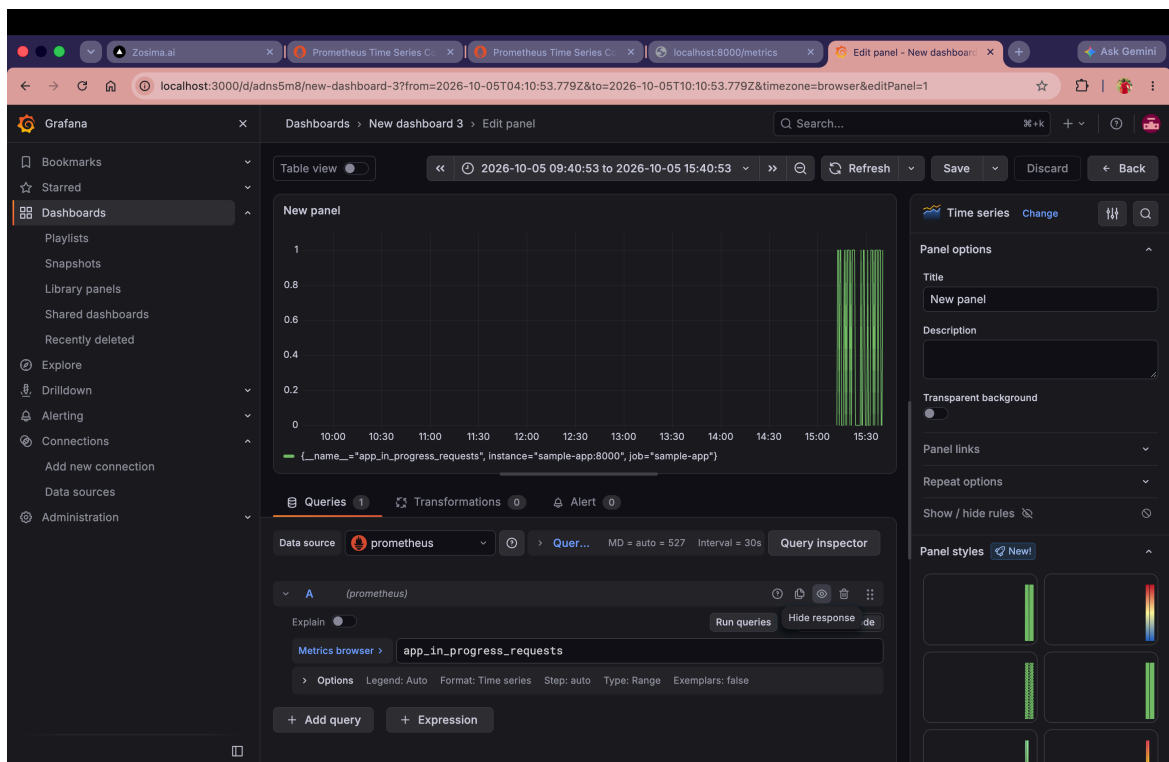
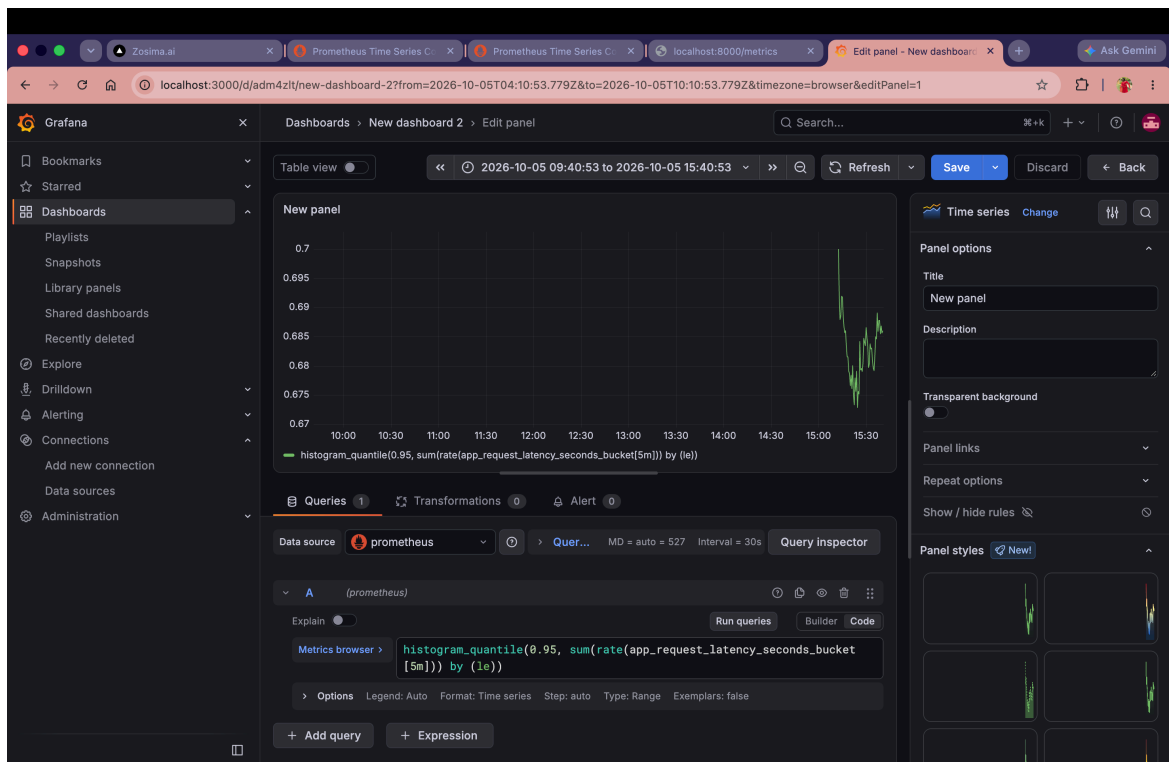
Build a Dashboard

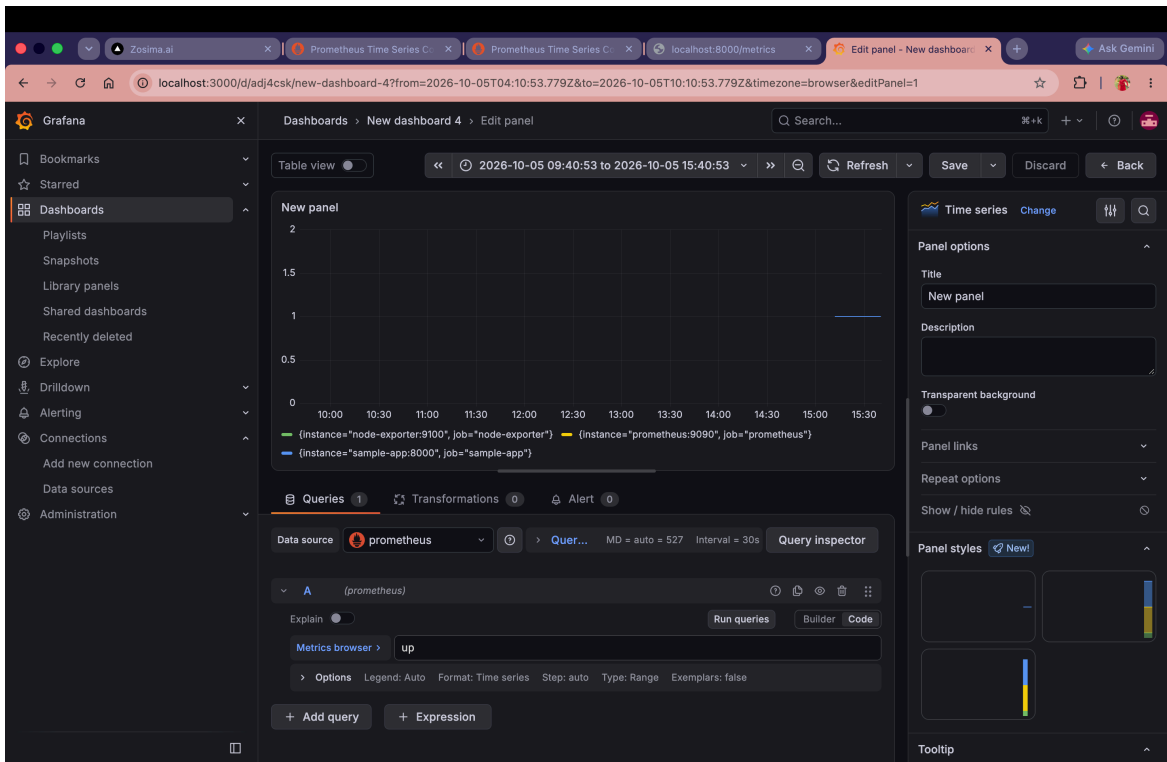
Panel	Query
Request rate	<code>sum(rate(app_requests_total[1m])) by (endpoint)</code>
Error rate (%)	<code>100 * sum(rate(app_requests_total{status="500"}[1m])) / sum(rate(app_requests_total[1m]))</code>
p95 latency	<code>histogram_quantile(0.95, sum(rate(app_request_latency_seconds_bucket[5m])) by (le))</code>
In-progress requests	<code>app_in_progress_requests</code>
Targets up	<code>up (use a Stat panel)</code>

B. Create a custom dashboard for the sample app

Go to **Dashboards** → **New** → **New dashboard** → **Add visualization**, pick Prometheus, and add these panels:







```
✓ Volume monitoring-lab_grafana-data Created 0.0s
✓ Container sample-app Started 0.2s
✓ Container prometheus Started 0.3s
✓ Container node-exporter Started 0.1s
✓ Container grafana Started 0.3s
NAME IMAGE COMMAND SERVICE CREATED STATUS PORTS
grafana grafana/grafana:latest "/run.sh" grafana Less than a second ago Up Less than a second 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
node-exporter prom/node-exporter:latest "/bin/node_exporter" node-exporter Less than a second ago Up Less than a second 0.0.0.0:9100->9100/tcp, [::]:9100->9100/tcp
prometheus prom/prometheus:latest "/bin/prometheus -c..." prometheus Less than a second ago Up Less than a second 0.0.0.0:9090->9090/tcp, [::]:9090->9090/tcp
sample-app monitoring-lab-sample-app "python -u /app.py" sample-app Less than a second ago Up Less than a second 0.0.0.0:8000->8000/tcp, [::]:8000->8000/tcp
anushkaberlia@Anushkas-MacBook-Air-2 monitoring-lab % cd ~/monitoring-lab
anushkaberlia@Anushkas-MacBook-Air-2 monitoring-lab % cat > alerts.yml << 'EOF'
groups:
- name: app-alerts
  rules:
  - alert: HighErrorRate
    expr: 100 * sum(rate(app_requests_total(status="500")[1m])) / sum(rate(app_requests_total[1m])) > 5
    for: 1m
    labels:
      severity: warning
    annotations:
      summary: "Error rate above 5%"
  EOF
anushkaberlia@Anushkas-MacBook-Air-2 monitoring-lab % cat > prometheus.yml << 'EOF'
global:
  scrape_interval: 5s

rule_files:
- alerts.yml

scrape_configs:
- job_name: prometheus
  static_configs:
  - targets: ['prometheus:9090']

- job_name: node-exporter
  static_configs:
  - targets: ['node-exporter:9100']

- job_name: sample-app
  static_configs:
  - targets: ['sample-app:8000']
EOF
anushkaberlia@Anushkas-MacBook-Air-2 monitoring-lab % volumes:
- ./prometheus.yml:/etc/prometheus/prometheus.yml
- ./alerts.yml:/etc/prometheus/alerts.yml
zsh: command not found: volumes:
zsh: no such file or directory: ./prometheus.yml:/etc/prometheus/prometheus.yml
zsh: no such file or directory: ./alerts.yml:/etc/prometheus/alerts.yml
anushkaberlia@Anushkas-MacBook-Air-2 monitoring-lab % docker compose up -d --force-recreate prometheus
[*] up 1/1
✓ Container prometheus Started
anushkaberlia@Anushkas-MacBook-Air-2 monitoring-lab % docker compose down # stops containers, keeps your Grafana dashboards
docker compose down -v # stops containers and deletes Grafana data too
no such service: #

What's next:
Debug this Compose error with Gordon → docker ai "help me fix this compose error"
no such service: #
```